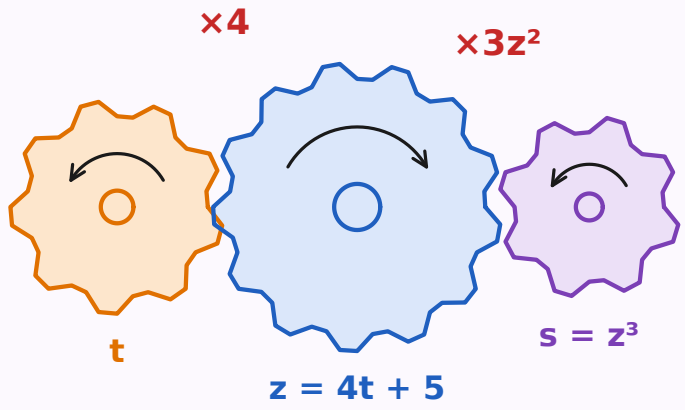
Chapter 35

Chain, product and quotient rules

rates multiply, one layer at a time



Turn the first gear once: the middle one turns 4 times as fast, the last one $3z^2$ times as fast as the middle. Altogether $4 \times 3z^2$. Rates multiply.

Chapter 34's rules handle one simple term at a time. Real formulas are built from pieces: one thing **inside** another, two things **multiplied**, one thing **divided** by another. Each way of building gets its own rule and its own picture:

- **chain rule**: gears turning gears (and nesting dolls)
- **product rule**: a rectangle growing on two sides
- **quotient rule**: a product wearing a disguise
- **implicit differentiation**: every y wears a luggage tag

Goes with **Lecture 15** (higher-order derivatives, functions of functions, chain rule, product rule, quotient rule, options, implicit differentiation).

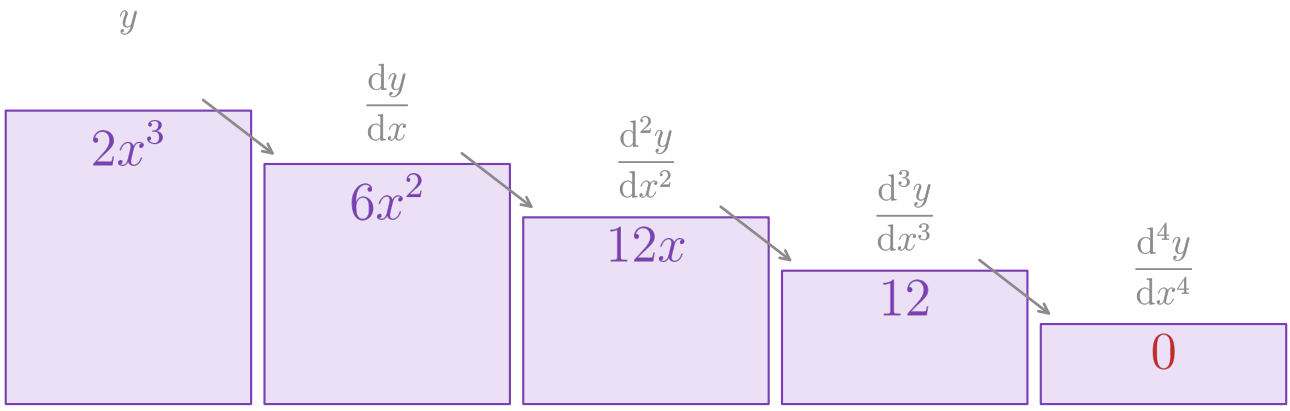
Differentiate again, and again

RULE: higher derivatives

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$: differentiate $\frac{\mathrm{d}y}{\mathrm{d}x}$ once more. $\frac{\mathrm{d}^3y}{\mathrm{d}x^3}$: once more again, and so on. Also written $f''(x)$, $f'''(x)$.

Use the same rules every time; rewrite $\frac{1}{x}$ as x^{-1} first and watch the signs of negative powers.

You can differentiate a derivative. Lecture 15: velocity differentiated is acceleration; acceleration differentiated says how the acceleration changes. For $y = 2x^3$ each differentiation walks one step down the stairs:



Powers of x eventually vanish; sine and cosine cycle for ever (Chapter 34's wheel). Lecture 15's examples:

$$y = 4x^3 - 2x^2 + \frac{1}{x} + 2 \longleftarrow 1/x \text{ is } x^{-1}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 12x^2 - 4x - \frac{1}{x^2} \longleftarrow x^{-1} \to -1x^{-2}$$

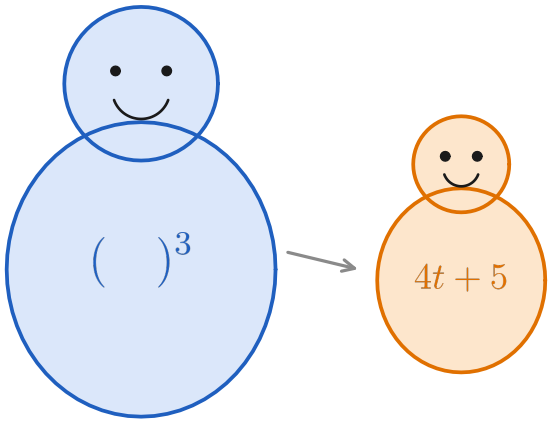
$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 24x - 4 + \frac{2}{x^3} \longleftarrow -x^{-2} \to +2x^{-3}$$

Slide slip: Lecture 15 writes $+\frac{1}{x^3}$ in the second derivative. The power rule on $-x^{-2}$ gives $(-1)(-2)x^{-3} = +2x^{-3}$, so it's $\frac{2}{x^3}$.

$$y = \sin x + 2x^3 \;\Rightarrow\; \frac{\mathrm{d}y}{\mathrm{d}x} = \cos x + 6x^2 \;\Rightarrow\; \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -\sin x + 12x$$

A function inside a function

Lecture 15: $s = (4t + 5)^3$. It's one function (**cube it**) wrapped around another ($4t + 5$), like nesting dolls:



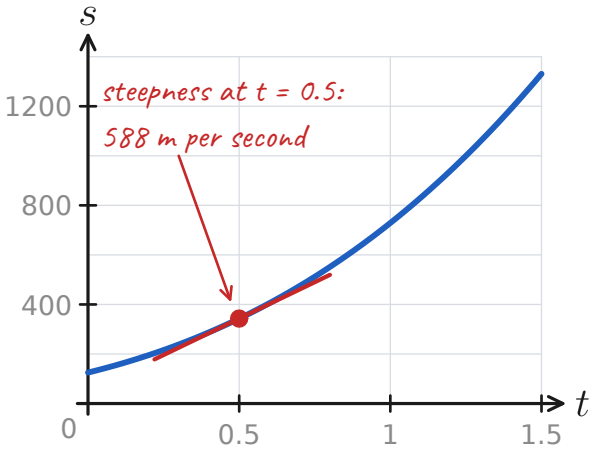
outside doll: "cube whatever is inside"
inside doll: 4t + 5

(Chapter 18 called this a composite function.)

So what are we trying to find? **Nothing here needs solving.** $s = (4t + 5)^3$ is a finished recipe: pick a time t and it tells you the distance s. Think of t as seconds and s as metres travelled. Lecture 15's question is a different one: **how fast is s changing?** That speed is $\frac{ds}{dt}$, Chapter 34's gradient (how steep the graph is).

t (s)	s (m)	jump
0	125	
		+218
0.5	343	
		+386
1	729	
		+602
1.5	1,331	

the jumps keep growing: s speeds up, so the speed depends on t



So the goal is a **new formula**, $\frac{ds}{dt}$: a speedometer. Feed in any t and it gives the speed at that moment. We'll get $\frac{ds}{dt} = 12(4t + 5)^2$, so at $t = 0.5$ the speed is $12 \times 7^2 = 588$, the slope of the red line.

The long way (what Lecture 15 does first): multiply it all out, then differentiate.

$$(4t + 5)^3 = (4t + 5)(16t^2 + 40t + 25) \quad \longleftarrow \textit{square first}$$

$$= 64t^3 + 240t^2 + 300t + 125$$

$$\frac{\mathrm{d}s}{\mathrm{d}t} = 192t^2 + 480t + 300 = 12(16t^2 + 40t + 25) = 12(4t+5)^2$$

take out 12, spot the square

Six lines, and imagine a power of 10! Look at the answer though: $12(4t+5)^2 = 3(4t+5)^2 \times 4$. That's the outside doll differentiated ($3(\ )^2$) times the inside doll differentiated (4). That's the shortcut.

The chain rule: gears

RULE: the chain rule

Use it for a **function of a function**: something with an inside chunk, like $(4t+5)^3$, e^{x^2}, $\ln(3x+1)$, $\sqrt{1+x^3}$, $\sin(2x)$.

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}u} \times \frac{\mathrm{d}u}{\mathrm{d}x}$, where u is the inside: **derivative of the outside** (inside left alone) **× derivative of the inside**.

PROCEDURE: chain rule step by step

1. Name the inside chunk u (any spare letter: z, u...).
2. Write y in terms of u alone.
3. Differentiate the outside with respect to u, and the inside with respect to x.
4. Multiply the two.
5. Put the inside back in place of u and tidy.

Name the inside z. Then t drives z, and z drives s, like the gears on the title page. How fast s changes with t is the two rates **multiplied**:

$$\frac{\mathrm{d}s}{\mathrm{d}t} = \frac{\mathrm{d}s}{\mathrm{d}z} \times \frac{\mathrm{d}z}{\mathrm{d}t}$$ ← *outside rate × inside rate*

(It looks as if the $\mathrm{d}z$'s cancel. They don't really, since these aren't fractions, but it's a handy way to remember it.)

Five small steps for $s = (4t+5)^3$:

$z = 4t + 5$ ← *1. name the inside doll*

$$s = z^3 \quad \longleftarrow \text{2. the outside doll}$$

$$\frac{\mathrm{d}s}{\mathrm{d}z} = 3z^2, \qquad \frac{\mathrm{d}z}{\mathrm{d}t} = 4 \quad \longleftarrow \text{3. differentiate each doll}$$

$$\frac{\mathrm{d}s}{\mathrm{d}t} = 3z^2 \times 4 = 12z^2 \quad \longleftarrow \text{4. multiply the rates}$$

$$\frac{\mathrm{d}s}{\mathrm{d}t} = 12(4t+5)^2 \quad \longleftarrow \text{5. put the inside back}$$

Same answer as the six-line slog. Lecture 15's second example: $z = 2(3t^3 + 2t^2)^4$.

$$u = 3t^3 + 2t^2, \qquad z = 2u^4 \quad \longleftarrow \text{name the inside}$$

$$\frac{\mathrm{d}z}{\mathrm{d}u} = 8u^3, \qquad \frac{\mathrm{d}u}{\mathrm{d}t} = 9t^2 + 4t \quad \longleftarrow \text{each doll}$$

$$\frac{\mathrm{d}z}{\mathrm{d}t} = 8(9t^2 + 4t)(3t^3 + 2t^2)^3 \quad \longleftarrow \text{multiply, put u back}$$

Lecture 15's quick version for brackets: bring the power down, knock one off the power, then multiply by the derivative of what's in the bracket. And the table in Chapter 34 was secretly using the chain rule all along: the k in $e^{kx} \to ke^{kx}$ is the inside's rate.

$$\frac{\mathrm{d}}{\mathrm{d}x}\sin(x^2) = \cos(x^2) \times 2x \quad \longleftarrow \text{outside: sin} \to \text{cos; inside: } x^2 \to 2x$$

The chain rule with e, ln and roots

These patterns come up constantly. Each is the chain rule with $g(x)$ as the inside and $g'(x)$ as its derivative:

outside	derivative	example
$(g(x))^n$	$n(g(x))^{n-1} \times g'(x)$	$(2x+1)^5 \to 5(2x+1)^4 \times 2$
$e^{g(x)}$	$g'(x)\,e^{g(x)}$: the exponential copies itself, times the inside's derivative	$e^{x^2} \to 2x\,e^{x^2}$
$\ln g(x)$	$\dfrac{g'(x)}{g(x)}$: inside's derivative over the inside	$\ln(3x+1) \to \dfrac{3}{3x+1}$
$\sqrt{g(x)} = g(x)^{\frac{1}{2}}$	$\frac{1}{2}g(x)^{-\frac{1}{2}}g'(x) = \dfrac{g'(x)}{2\sqrt{g(x)}}$	$\sqrt{x^2+1} \to \dfrac{2x}{2\sqrt{x^2+1}}$
$\sin g(x)$, $\cos g(x)$	$g'(x)\cos g(x)$, $-g'(x)\sin g(x)$	$\cos(x^2) \to -2x\sin(x^2)$

Worked examples. Name g, find g', then use the pattern:

$$y = e^{x^2+3x}: \quad g' = 2x+3, \quad \frac{\mathrm{d}y}{\mathrm{d}x} = (2x+3)\,e^{x^2+3x}$$

e copies itself × g'

$$y = \ln(x^2+4): \quad \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{2x}{x^2+4}$$ ← *g' over g*

$$y = \ln(5x): \quad \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{5}{5x} = \frac{1}{x}$$ ← *same as ln x! (ln 5x = ln 5 + ln x)*

$$y = \sqrt{4x+1} = (4x+1)^{\frac{1}{2}}$$ ← *√ is the power ½*

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \tfrac{1}{2}(4x+1)^{-\frac{1}{2}} \times 4 = \frac{2}{\sqrt{4x+1}}$$ ← *× 4 from the inside*

$$y = \frac{3}{(2x-1)^2} = 3(2x-1)^{-2}$$ ← rewrite as a power first

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -6(2x-1)^{-3} \times 2 = -\frac{12}{(2x-1)^3}$$ ← then chain: inside 2x − 1 → 2

✗ Common mistakes

✗ $\frac{\mathrm{d}}{\mathrm{d}x} e^{x^2} = e^{x^2}$

multiply by the inside's derivative

✓ $\frac{\mathrm{d}}{\mathrm{d}x} e^{x^2} = 2x\, e^{x^2}$

✗ $\frac{\mathrm{d}}{\mathrm{d}x} \ln(x^2+4) = \frac{1}{x^2+4}$

the top is g', not 1

✓ $\frac{\mathrm{d}}{\mathrm{d}x} \ln(x^2+4) = \frac{2x}{x^2+4}$

✗ $\frac{\mathrm{d}}{\mathrm{d}x}(2x+1)^5 = 5(2x+1)^4$

× 2 from the inside

✓ $\frac{\mathrm{d}}{\mathrm{d}x}(2x+1)^5 = 10(2x+1)^4$

The product rule: a rectangle growing on two sides

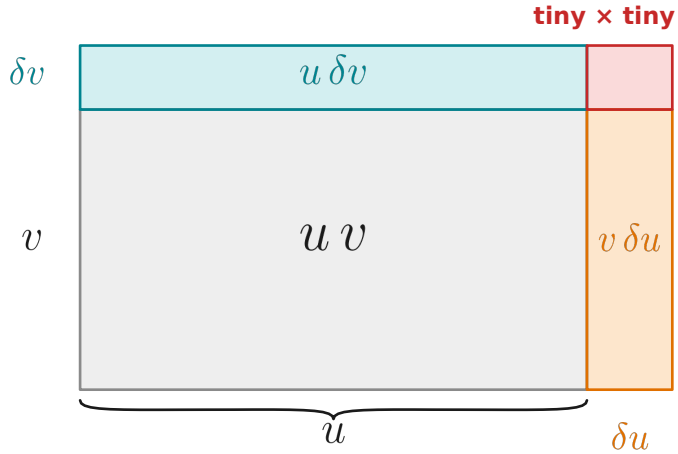
RULE: the product rule

For $y = uv$, where **both** u and v contain x: $\frac{dy}{dx} = v\frac{du}{dx} + u\frac{dv}{dx}$. In words: (second × derivative of first) + (first × derivative of second).

Steps: write down u, v, u', v' (use the chain rule for these if needed), substitute, then simplify, often by taking out a common factor.

A number times a function is not a product: $\frac{d}{dx}(5\sin x) = 5\cos x$. The number just waits in front.

$y = u \times v$, where u and v both depend on x. Picture y as the **area** of a rectangle with sides u and v. Nudge x a little: both sides grow.



extra area =
v δu (orange) + u δv (teal)
+ δu δv (red corner)

Shrink the nudge: the red corner is tiny × tiny, so it vanishes. What's left:

$$\frac{dy}{dx} = v\frac{du}{dx} + u\frac{dv}{dx}$$ ← *each side's growth × the other side*

Test it where you can check (Lecture 15): $y = (3x+1)(5x-2)$.

expand first: $y = 15x^2 - x - 2$, $\quad \frac{dy}{dx} = 30x - 1$

the old way

$u = 3x+1,\ v = 5x-2,\ \frac{du}{dx} = 3,\ \frac{dv}{dx} = 5$

$$\frac{dy}{dx} = (5x-2)\times 3 + (3x+1)\times 5 = 15x - 6 + 15x + 5 = 30x - 1$$

✓ *the same*

Where you have no choice: $y = x^3 \ln x$ can't be expanded.

$$u = x^3, \quad v = \ln x, \quad \frac{\mathrm{d}u}{\mathrm{d}x} = 3x^2, \quad \frac{\mathrm{d}v}{\mathrm{d}x} = \frac{1}{x}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \ln x \times 3x^2 + x^3 \times \frac{1}{x} = 3x^2 \ln x + x^2 = x^2(1 + 3\ln x)$$

Product rule: Lecture 15's harder ones

Your turn slide: $y = x^{-2}(4 + 3x^{-3})$.

$$u = x^{-2}, \; v = 4 + 3x^{-3}, \; \frac{\mathrm{d}u}{\mathrm{d}x} = -2x^{-3}, \; \frac{\mathrm{d}v}{\mathrm{d}x} = -9x^{-4}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = (4 + 3x^{-3})(-2x^{-3}) + x^{-2}(-9x^{-4}) \longleftarrow v\,u' + u\,v'$$

$$= -8x^{-3} - 6x^{-6} - 9x^{-6} = -8x^{-3} - 15x^{-6} = -\frac{8x^3 + 15}{x^6}$$

add powers (Chapter 9)

Combined example: $y = (x + 3)^2(4x + 1)^3$. Each factor is itself a nesting doll, so the chain rule works **inside** the product rule:

$$u = (x + 3)^2 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2(x + 3) \times 1 \longleftarrow \text{chain}$$

$$v = (4x + 1)^3 \Rightarrow \frac{\mathrm{d}v}{\mathrm{d}x} = 3(4x + 1)^2 \times 4 = 12(4x + 1)^2$$

chain

$$\frac{\mathrm{d}y}{\mathrm{d}x} = (4x + 1)^3 \cdot 2(x + 3) + (x + 3)^2 \cdot 12(4x + 1)^2$$

product rule

$$= 320x^4 + 1728x^3 + 2628x^2 + 1010x + 114 \longleftarrow \text{multiplied out}$$

Slide slip: Lecture 15's multiplied-out answer starts $340x^4$; it's $320x^4$ ($2 \times 64 + 12 \times 16 = 128 + 192 = 320$).

Tidier: take out the common factors $2(x+3)(4x+1)^2$, leaving $\frac{dy}{dx} = 2(x+3)(4x+1)^2(10x+19)$.

The quotient rule: a product in disguise

RULE: the quotient rule

For $y = \frac{u}{v}$: $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{v\frac{\mathrm{d}u}{\mathrm{d}x} - u\frac{\mathrm{d}v}{\mathrm{d}x}}{v^2}$. The order on top matters because of the minus: **bottom × top′ − top × bottom′**.

Do not expand v^2 on the bottom; simplify the top only.

If the bottom is just a power of x (like $\frac{x^2+1}{x}$), split the fraction and use the power rule instead. If the bottom is a number, it is just a constant multiplier.

A fraction $\frac{u}{v}$ is really $u \times v^{-1}$. So you don't need anything new: use the product rule, with the chain rule for v^{-1}:

$$y = u\,v^{-1}$$ ← dividing = multiplying by the reciprocal

$$\frac{\mathrm{d}y}{\mathrm{d}x} = v^{-1}\frac{\mathrm{d}u}{\mathrm{d}x} + u\cdot(-1)v^{-2}\frac{\mathrm{d}v}{\mathrm{d}x}$$ ← product rule; chain on v^{-1}

$$= \frac{v\frac{\mathrm{d}u}{\mathrm{d}x} - u\frac{\mathrm{d}v}{\mathrm{d}x}}{v^2}$$ ← over a common denominator v^2

Memory rhyme: **"low d-high minus high d-low, over the square of what's below"**. The order matters because of the minus.

Lecture 15: $y = \frac{x-1}{x+2}$, so $u = x-1$, $v = x+2$, and both derivatives are 1.

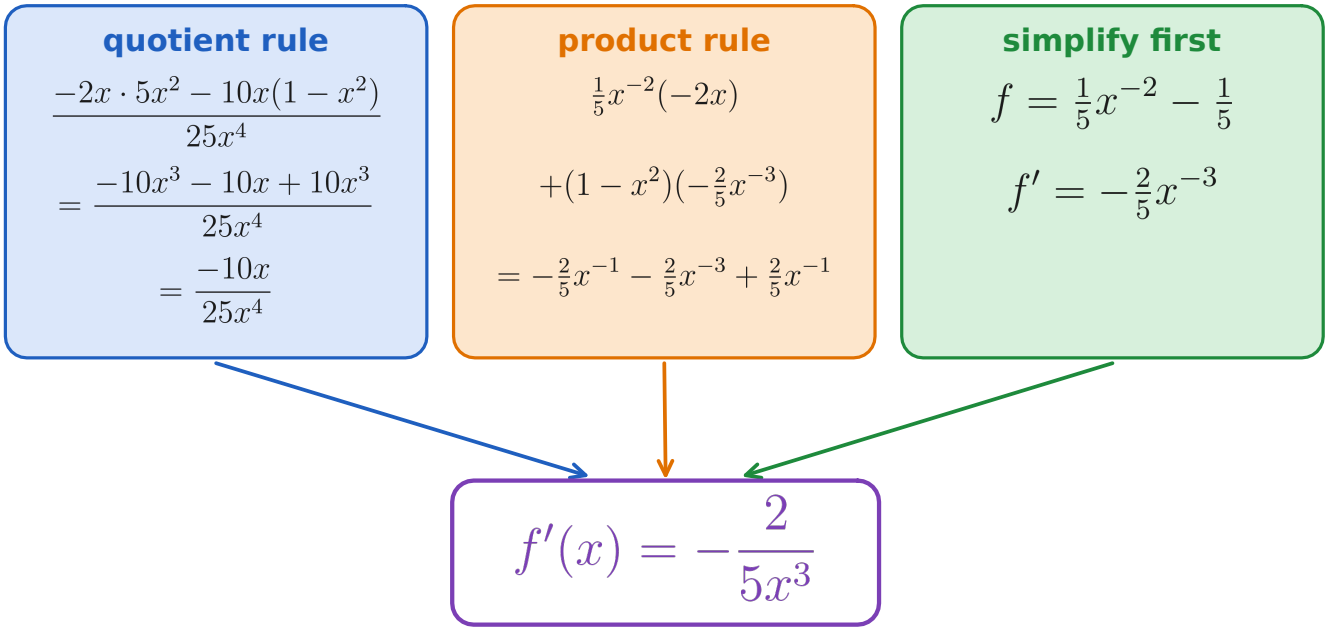
$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{(x+2)(1) - (x-1)(1)}{(x+2)^2} = \frac{3}{(x+2)^2}$$

Harder: $y = \frac{e^{-3t}}{t^2+1}$, with $u = e^{-3t}$, $u' = -3e^{-3t}$, $v = t^2+1$, $v' = 2t$.

$$\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{(t^2+1)(-3e^{-3t}) - e^{-3t}(2t)}{(t^2+1)^2} = \frac{-(3t^2+2t+3)\,e^{-3t}}{(t^2+1)^2}$$

Three roads to the same answer

Lecture 15's "options": $f(x) = \frac{1-x^2}{5x^2}$. The quotient rule works, the product rule works, and a third road (simplify first) is the shortest:



Moral: **tidy up before you differentiate**. Splitting the fraction ($\frac{1}{5x^2} - \frac{x^2}{5x^2}$) turned a quotient into two plain powers.

Implicit differentiation: every y wears a tag

PROCEDURE: implicit differentiation

1. Differentiate **every term** of the equation with respect to x, both sides.
2. An x term differentiates as usual. A y term: differentiate as if y were x, then multiply by $\frac{dy}{dx}$ (chain rule). A number on its own gives 0.
3. Collect every $\frac{dy}{dx}$ term on one side and everything else on the other.
4. Factorise out $\frac{dy}{dx}$ and divide.
5. For a gradient at a point you need **both** x and y. If only x is given, put it into the **original** equation and solve for y first.

Explicit: $y =$ (something with x). **Implicit**: x and y are tangled together, like $x^3 + 3y^2 - 4x + y = 11$, and you can't easily get y on its own.

The trick: y is secretly a function of x (an inside doll!). So when you differentiate a y term with respect to x, the chain rule sticks a **tag** on it: differentiate as if y were x, then multiply by $\frac{dy}{dx}$.

$$\underset{\downarrow}{x^3} \quad \underset{\downarrow}{+3y^2} \quad \underset{\downarrow}{-4x} \quad \underset{\downarrow}{+y} \quad = 11$$
$$3x^2 \quad +6y \cdot \tfrac{dy}{dx} \quad -4 \quad +1 \cdot \tfrac{dy}{dx} \quad = 0$$

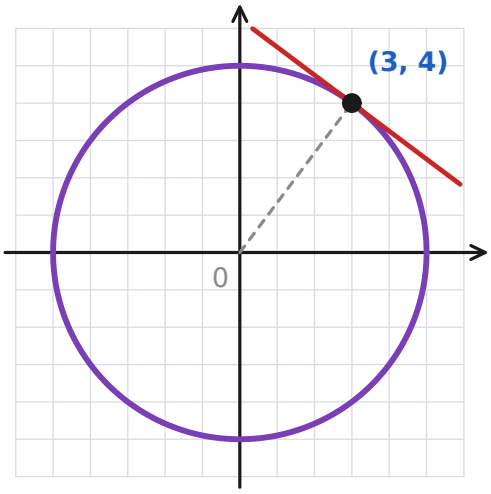
$$3x^2 + 6y\frac{dy}{dx} - 4 + \frac{dy}{dx} = 0$$ ← the tagged terms hold dy/dx

$$\frac{dy}{dx}(6y + 1) = 4 - 3x^2$$ ← tagged terms on one side; factor (Chapter 12)

$$\frac{dy}{dx} = \frac{4 - 3x^2}{6y + 1}$$ ← ÷ (6y + 1)

Notice the answer has y in it: the gradient depends on **where** you are on the curve, which needs both coordinates.

A picture you can check: the circle $x^2 + y^2 = 25$ (Chapter 28).



$$2x + 2y\frac{\mathrm{d}y}{\mathrm{d}x} = 0$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{x}{y}$$

$$\text{at } (3,4): \ -\tfrac{3}{4}$$

The radius has gradient 4/3, the tangent −3/4: perpendicular, as Chapter 20 promised (they multiply to −1).

When only x is given. Find the gradient of $y^3 + x^2 = 9$ at the point where $x = 1$.

$$y^3 + 1^2 = 9 \ \Rightarrow \ y^3 = 8 \ \Rightarrow \ y = 2$$ ← *step 5 first: the point is (1, 2)*

$$3y^2\frac{\mathrm{d}y}{\mathrm{d}x} + 2x = 0$$ ← *steps 1–2: y^3 gets a tag*

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{2x}{3y^2}$$ ← *steps 3–4*

$$\text{at } (1,2): \ -\frac{2(1)}{3(2)^2} = -\frac{2}{12} = -\frac{1}{6}$$

Which rule? Sort by how it's built

PROCEDURE: choosing the rule

1. **Simplify first** if you can: split a fraction over a single power of x, expand a small bracket, use log laws ($\ln x^3 = 3\ln x$). Often no special rule is needed afterwards.
2. A **number times a function** ($5\sin x$, $3e^{2x}$, $\frac{x^2}{4}$): keep the number in front. **Not** a product-rule case.
3. A **function of a function** (an inside chunk): **chain rule**.
4. Two functions of x **multiplied**: **product rule**.
5. One function of x **divided** by another: **quotient rule** (or rewrite as $u \times v^{-1}$ and use the product rule).
6. x and y mixed in one equation: **implicit differentiation**.
7. Rules can nest: inside a product, each of u and v may need the chain rule.

Ask: what is the **last** thing you'd do if you worked it out on a calculator? That's the outer structure, and it picks the rule.

something inside something: CHAIN	two x-things multiplied: PRODUCT	one divided by another: QUOTIENT	x and y tangled: IMPLICIT
$(2x+1)^5$	$x^2 \sin x$	$\frac{x+1}{x-1}$	$x^2 + y^2 = 25$
e^{3x^2}	xe^x	$\frac{\sin x}{x}$	$x^3 + 3y^2 = 11$
$\sin(x^2)$	$(x+1)^3(x-2)$	$\frac{e^x}{x^2+1}$	$y^3 + x = y$
$\sqrt{1+x^3}$	$x \ln x$		

Sums and differences are never a problem: split them up and do each term with its own rule.

✗ Common mistakes

✗ $\frac{\mathrm{d}}{\mathrm{d}x}\ln(5x) = \frac{1}{5x}$

ln of an inside chunk: g' over g

✓ $\frac{\mathrm{d}}{\mathrm{d}x}\ln(5x) = \frac{5}{5x} = \frac{1}{x}$

✗ $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(x^2 \sin x\right) = 2x \cos x$ ✓ $2x \sin x + x^2 \cos x$

a product is NOT the product of the derivatives

✗ $\dfrac{\mathrm{d}}{\mathrm{d}x}\dfrac{u}{v} = \dfrac{u'}{v'}$ ✓ $\dfrac{vu' - uv'}{v^2}$

a quotient is NOT the quotient of the derivatives

Exam combinations

Exam questions mix this chapter's rules with Chapter 34's jobs: gradient at a point, stationary points, tangents. Pick the rule, differentiate, then carry on exactly as in Chapter 34.

Stationary point of $y = xe^{-x}$. Product rule, with the chain rule for e^{-x}:

$$u = x, \quad v = e^{-x}, \quad u' = 1, \quad v' = -e^{-x}$$ ← chain: inside −x → −1

$$\frac{\mathrm{d}y}{\mathrm{d}x} = e^{-x} \cdot 1 + x \cdot \left(-e^{-x}\right) = e^{-x}(1 - x)$$ ← v u′ + u v′, take out e^{-x}

$$e^{-x}(1 - x) = 0 \;\Rightarrow\; x = 1$$ ← e^{-x} is never 0, so 1 − x = 0

$$y = 1 \cdot e^{-1} = \frac{1}{e} \approx 0.368$$ ← y from the ORIGINAL equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = e^{-x}(x - 2); \quad x = 1 : \; -e^{-1} < 0$$ ← product rule again: maximum

So $\left(1, \frac{1}{e}\right)$ is a **maximum**.

Gradient at a point after the quotient rule: $y = \frac{2x}{x+1}$ at $x = 1$.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{(x+1) \cdot 2 - 2x \cdot 1}{(x+1)^2} = \frac{2}{(x+1)^2}$$ ← simplify the top

$$x = 1 : \quad \frac{2}{2^2} = \frac{1}{2}$$ ← then substitute

Gradient at a point after the chain rule: $y = \sqrt{2x+5}$ at $x = 2$.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{2}{2\sqrt{2x+5}} = \frac{1}{\sqrt{2x+5}}; \quad x = 2: \ \frac{1}{\sqrt{9}} = \frac{1}{3}$$

Find the unknown: $y = e^{kx}$ has gradient 6 at $x = 0$. Find k.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = ke^{kx}; \quad x = 0: \ ke^{0} = k = 6 \longleftarrow e^{0} = 1$$

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Higher derivatives	$\dfrac{d^2y}{dx^2} = \dfrac{d}{dx}\left(\dfrac{dy}{dx}\right)$ differentiate again
Chain rule	$\dfrac{dy}{dx} = \dfrac{dy}{du} \times \dfrac{du}{dx}$ outside's derivative × inside's derivative
Chain: powers	$\dfrac{d}{dx}\big(g(x)\big)^n = n\big(g(x)\big)^{n-1} g'(x)$
Chain: e and ln	$\dfrac{d}{dx}e^{g(x)} = g'(x)e^{g(x)}$, $\qquad \dfrac{d}{dx}\ln g(x) = \dfrac{g'(x)}{g(x)}$
Chain: roots	$\dfrac{d}{dx}\sqrt{g(x)} = \dfrac{g'(x)}{2\sqrt{g(x)}}$
Product rule	$\dfrac{d(uv)}{dx} = v\dfrac{du}{dx} + u\dfrac{dv}{dx}$ only when both factors contain x
Quotient rule	$\dfrac{d}{dx}\left(\dfrac{u}{v}\right) = \dfrac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$ bottom × top′ − top × bottom′, over bottom²
Constant multiple	$\dfrac{d}{dx}\big(kf(x)\big) = kf'(x)$ not a product
Implicit	differentiate every term; each y term gets $\times\frac{dy}{dx}$; collect and factorise. Need y? Use the original equation
Which rule?	simplify first; inside chunk: chain; multiplied: product; divided: quotient; x and y mixed: implicit

Your turn

Name the rule before you start. Answers in grey.

Higher derivatives

1. Find d^2y/dx^2 for $y = x^4 - 3x^2 + 5$.
2. Find d^3y/dx^3 for $y = \cos x$.

1. $12x^2 - 6$ 2. $\sin x$

Chain rule

3. $y = (2x + 1)^5$
4. $y = e^{(3x^2)}$
5. $y = \sqrt{(1 + x^3)}$

3. $10(2x + 1)^4$ 4. $6x\, e^{(3x^2)}$ 5. $3x^2 \div (2\sqrt{(1 + x^3)})$

Product rule

6. $y = x^2 \sin x$
7. $y = x\, e^x$

6. $2x \sin x + x^2 \cos x$ 7. $e^x + x\, e^x = (1 + x)e^x$

Quotient rule (or rewrite!)

8. $y = (x + 1)/(x - 1)$
9. $y = (x^2 + 1)/x$

8. $-2/(x - 1)^2$ 9. $1 - 1/x^2$ (easiest: $x + x^{-1}$ first)

Implicit

10. Find dy/dx for $x^2 + y^2 = 100$, and its value at $(6, 8)$.
11. Find dy/dx for $y^3 + x = y$.

10. $-x/y$; $-3/4$ 11. $1/(1 - 3y^2)$

Exam combinations

12. Find the stationary point of $y = x\, e^{(-2x)}$ and say what kind it is.
13. Find the gradient of $y = x^2/(x + 1)$ at $x = 1$.
14. $y = \ln(x^2 + 1)$: find dy/dx and its value at $x = 1$.

12. $dy/dx = e^{(-2x)}(1 - 2x) = 0$, $x = ½$; $(½, 1/(2e))$, maximum 13. $(x^2 + 2x)/(x + 1)^2 = 3/4$
14. $2x/(x^2 + 1)$; 1

Find the unknown

15. y = e^(ax) has gradient 6 at x = 0. Find a.
16. y = (ax + 1)⁴ has gradient 24 at x = 0. Find a.
17. y = ln(x² + k) has gradient ½ at x = 1. Find k.
18. y = x/(x + c) has gradient ¼ at x = 0 (c > 0). Find c.
19. (2, b) lies on x² + y³ = 12. Find b, then dy/dx there.

15. a = 6 (a e⁰ = 6) 16. a = 6 (4a(1)³ = 24) 17. k = 3 (2/(1 + k) = ½) 18. c = 4 (c/c² = 1/c = ¼) 19. b = 2 (4 + b³ = 12); dy/dx = −2x/(3y²) = −1/3